Differential Climate Impacts on Fishing Effort in Chilean Small Pelagic Fisheries

Online Appendix

Felipe J. Quezada-Escalona

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A Reduced-form stress tests and prior elicitation

This appendix documents the protocol used to elicit the prior densities on the climate shifters $(\rho_s^{SST}, \rho_s^{CHL})$ that enter the state-space specification of the Stock dynamics subsection of the main text, together with the identifiability diagnostic that motivates the recourse to a Bayesian likelihood rather than a direct maximum-likelihood treatment.

A.1 Deterministic hindcast and cross-variant fit

For each stock s we estimate a deterministic Schaefer hindcast over 2000–2024, $\hat{B}_{s,t+1} = \hat{B}_{s,t} + r_{s,t}\hat{B}_{s,t}(1-\hat{B}_{s,t}/K_s)-C_{s,t}$ with $r_{s,t} = r_s^0 \exp(\rho_s^{SST}(SST_{t-1}-\overline{SST})+\rho_s^{CHL}(\log CHL_{t-1}-\overline{\log CHL}))$, biological parameters $(r_s^0, K_s, B_{0,s})$ held at the centres of the IFOP/SPRFMO single-species priors, and shifters estimated by bounded least squares on $[-3,3]^2$ minimising median absolute percent error (MAPE). Four nested variants are reported: base ($\rho = 0$), +SST only, + logCHL only, and joint +SST+ logCHL.

No variant brings any of the three stocks below a 20% MAPE threshold. The two coastal stocks benefit from activating SST (anchoveta: 95.9% → 30.8%, sardine: 98.5% → 23.1%), with the joint

Table A.1: Median absolute percent error of the deterministic Schaefer hindcast across shifter variants.

Stock	Base	+ SST	+ log CHL	+ SST and log CHL
Anchoveta	95.9	30.8	95.1	26.6
Sardina común	98.5	23.1	98.5	20.5
Jack mackerel	44.2	44.3	49.1	48.5

specification delivering further marginal improvement. Jack mackerel hindcast errors remain in the 44–49% range across all variants, with no shifter distinguishably better than the base.

The joint-variant point estimates carry an internally coherent biological reading for the coastal stocks. Anchoveta: both shifters negative and similar in magnitude ($\rho^{SST} \approx -2.3$, $\rho^{CHL} \approx -2.3$). Sardina com'un: $\rho^{SST} \approx -2.0$ but $\rho^{CHL} \approx +2.1$ (opposite sign to anchoveta, consistent with the warm-water mesotrophic affinity that distinguishes sardine from anchoveta in the Humboldt Current biological literature). Jack mackerel: $\rho^{SST} \approx +0.6$ small and the ρ^{CHL} estimate binds at the lower boundary of the admissible box; we read the boundary solution as a failure of point identification rather than a finding of strong climate forcing.

A.2 Identifiability diagnostic and prior elicitation

Two diagnostics motivate the use of weakly informative priors rather than direct maximum-likelihood estimation. First, the two environmental forcings are statistically orthogonal over the 2000–2024 sample (Pearson $r = 0.030$, $p = 0.888$), so the cross-variant gap in hindcast error cannot be attributed to collinearity in the design matrix. Second, the jurel MLE binds at the boundary, indicating that the climate signal is not separately identifiable from process and observation noise absorbed into the residual under a deterministic single-species hindcast; the Bayesian state-space specification addresses both limitations within a single inferential framework.

Table A.2 reports the priors on $(\rho_s^{SST}, \rho_s^{CHL})$ used in the structural likelihood. Prior centres for the two coastal stocks are the joint MLEs rounded to one decimal place; jurel priors are centred at zero so that posterior identification is driven by the structural likelihood rather than by the boundary solution. All priors are independent normal with unit standard deviation; no hierarchical pooling is imposed because ρ^{CHL} switches sign between anchoveta and sardine.

This prior-elicitation strategy is a form of empirical Bayes: the prior centres for anchoveta and sardine are derived from the same 2000–2024 biomass series that enters the state-space likelihood, which trades pure data-prior independence for tractable identification under $N = 25$ annual observations. We adopt this trade-off deliberately for two reasons. First, the prior standard deviation $\sigma = 1$ is large enough that the posterior is free to move substantially: the posterior of ρ_s^{CHL} for anchoveta migrates to -3.65 (prior mean -2.3), more than one prior standard deviation from the centre, illustrating that the data does refine the prior when informative. Second, the alternative of centring all priors at zero would, under $N = 25$, leave the joint posterior of $(r_s^0, K_s, B_{0,s}, \rho_s^{SST}, \rho_s^{CHL})$ weakly identified or multimodal, producing posteriors that are uninformative for projections regardless of whether the underlying coupling is real or absent. The empirical-Bayes choice therefore makes the non-identification result for jack mackerel a substantive finding rather than a numerical artefact of weak posterior geometry.

Table A.2: Independent normal priors on the climate shifters used in the state-space likelihood.

Stock	$\mu_{\rho^{SST}}$	$\sigma_{\rho^{SST}}$	$\mu_{\rho^{CHL}}$	$\sigma_{\rho^{CHL}}$	Source
Anchoveta	-2.3	1.0	-2.3	1.0	Joint MLE
Sardina común	-2.0	1.0	+2.1	1.0	Joint MLE
Jack mackerel	0.0	1.0	0.0	1.0	Boundary solution; vague prior

B Predictive diagnostics

This appendix reports the nested predictive comparison referenced in the Stock dynamics subsection of the main text, comparing three state-space specifications: (i) species-independent dynamics without climate shifters (*ind*), (ii) the same with a cross-stock residual covariance Ω (*omega*), and (iii) the full model augmented with the climate shifter of Eq.~(1) of the main text (*full*). Two predictive metrics are reported: leave-one-out PSIS-LOO (Pareto-smoothed importance sampling) and one-step-ahead LFO (leave-future-out).

PSIS-LOO favours the climate-augmented specification with a positive ΔELPD of 14–18 relative to the climate-free specifications, indicating that the in-sample predictive performance improves when the climate shifter is included.

The one-step-ahead LFO comparison is essentially a tie across the three specifications, indicat-

Table B.1: PSIS-LOO comparison across state-space specifications.

Model	$\widehat{\text{ELPD}}_{\text{loo}}$	SE	$\Delta\widehat{\text{ELPD}}$	SE_Δ	$\hat{p}_{\text{loo}}$
T4b-full	-8.41	8.28	–	–	64.8
T4b-ind	-23.00	8.01	-14.59	4.93	62.4
T4b-omega	-26.43	9.16	-18.02	5.59	57.9

Note:
Δ ELPD is the pairwise difference relative to T4b-full (the leading model); the adjacent SE column reports the standard error of that difference. After moment matching, 65/68 observations satisfy Pareto-k <= 0.7 for T4b-full, 64/68 for T4b-ind, and 67/68 for T4b-omega. The remaining high-Pareto-k points are the two censored jack-mackerel years (2012, 2015), which appear in all three models and therefore do not affect the ranking.

Table B.2: One-step-ahead LFO comparison across state-space specifications.

Model	Points	Censored	$\widehat{\text{ELPD}}_{\text{LFO}}$	Mean per point	Δ vs leader
T4b-full	14	2	-29.10	-2.08	–
T4b-ind	14	2	-29.46	-2.10	-0.37
T4b-omega	14	2	-30.46	-2.18	-1.36

Note:
Sum of one-step predictive log densities over five origin years (2011, 2014, 2017, 2020, 2022) and three stocks (14 targets per model; two are censored). Each refit uses 4 chains × 1500 warmup + 1500 sampling iterations. All 15 refits satisfied R-hat <= 1.01 with at most one divergent transition (T4b-omega at the 2011 origin), which does not affect the reported ELPD.

ing that the climate-augmented specification does not deliver a detectable improvement in short-horizon forecast performance over the climate-free alternatives. We interpret this as a sample-size issue rather than a deficiency of the climate specification: the structural climate elasticities are identified on the full 25-year window, not on one-step-ahead increments. The posterior of the climate-augmented specification remains our primary object of interpretation in the main text.

C Posterior-predictive checks

This appendix reports posterior-predictive diagnostics of the full state-space specification: (i) the smoothed latent biomass versus the observed survey-based estimates, and (ii) the corresponding year-level residuals.

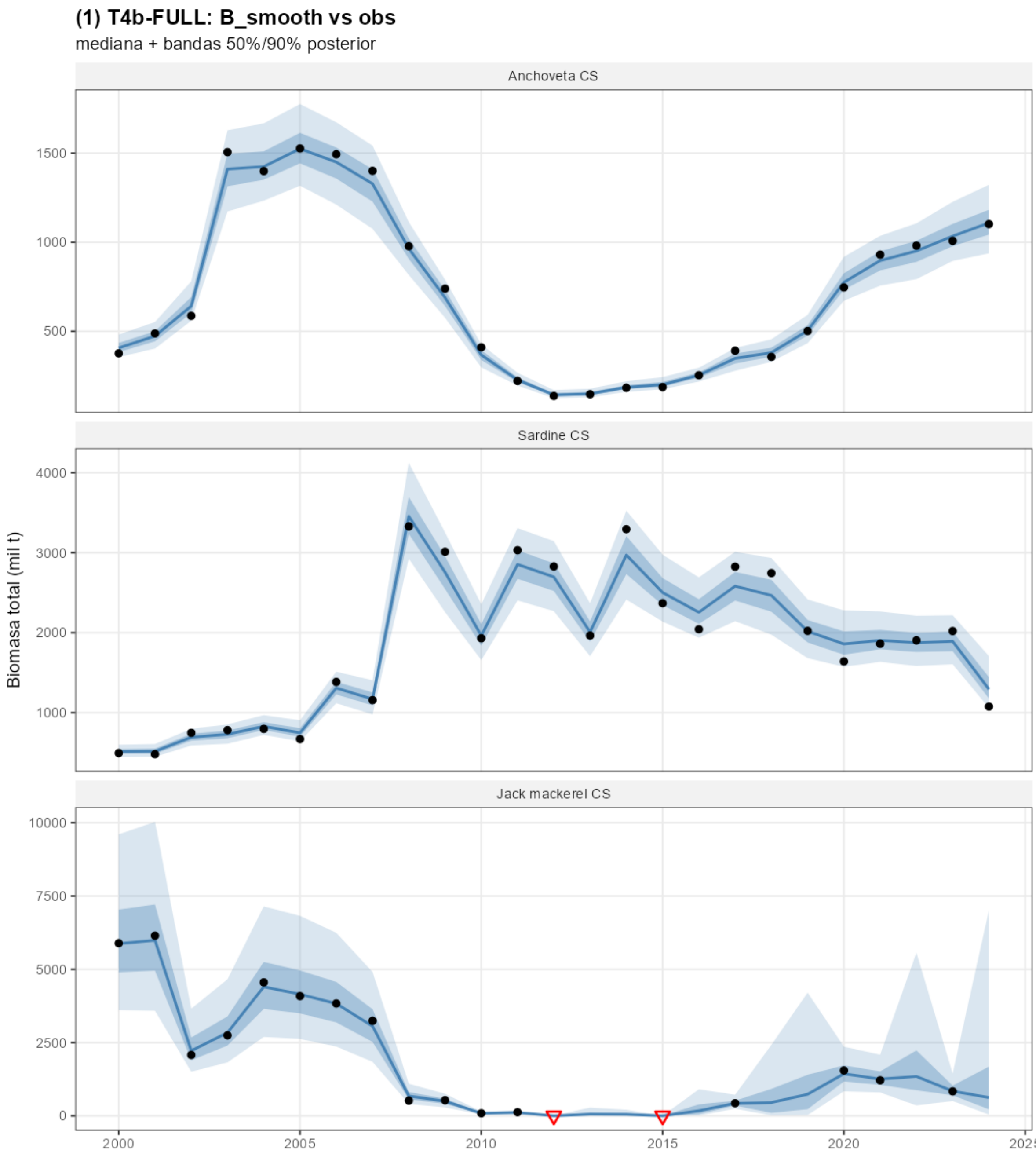

Figure C.1: Smoothed latent biomass $B_{i,t}$ of the full specification against survey-based SSB observations, by stock. Solid line: posterior median; shaded band: 90% credible interval. Open circles: official IFOP/SPRFMO point estimates; downward arrows: left-censored observations at the assessment detection limit (jurel 2012, 2015).

The smoothed band has a median width of approximately 20% of the posterior mean, tightening over years with direct observation and widening across the seven non-surveyed jurel years (latent biomass identified dynamically through the Schaefer transition and two censored observations in

2012 and 2015).

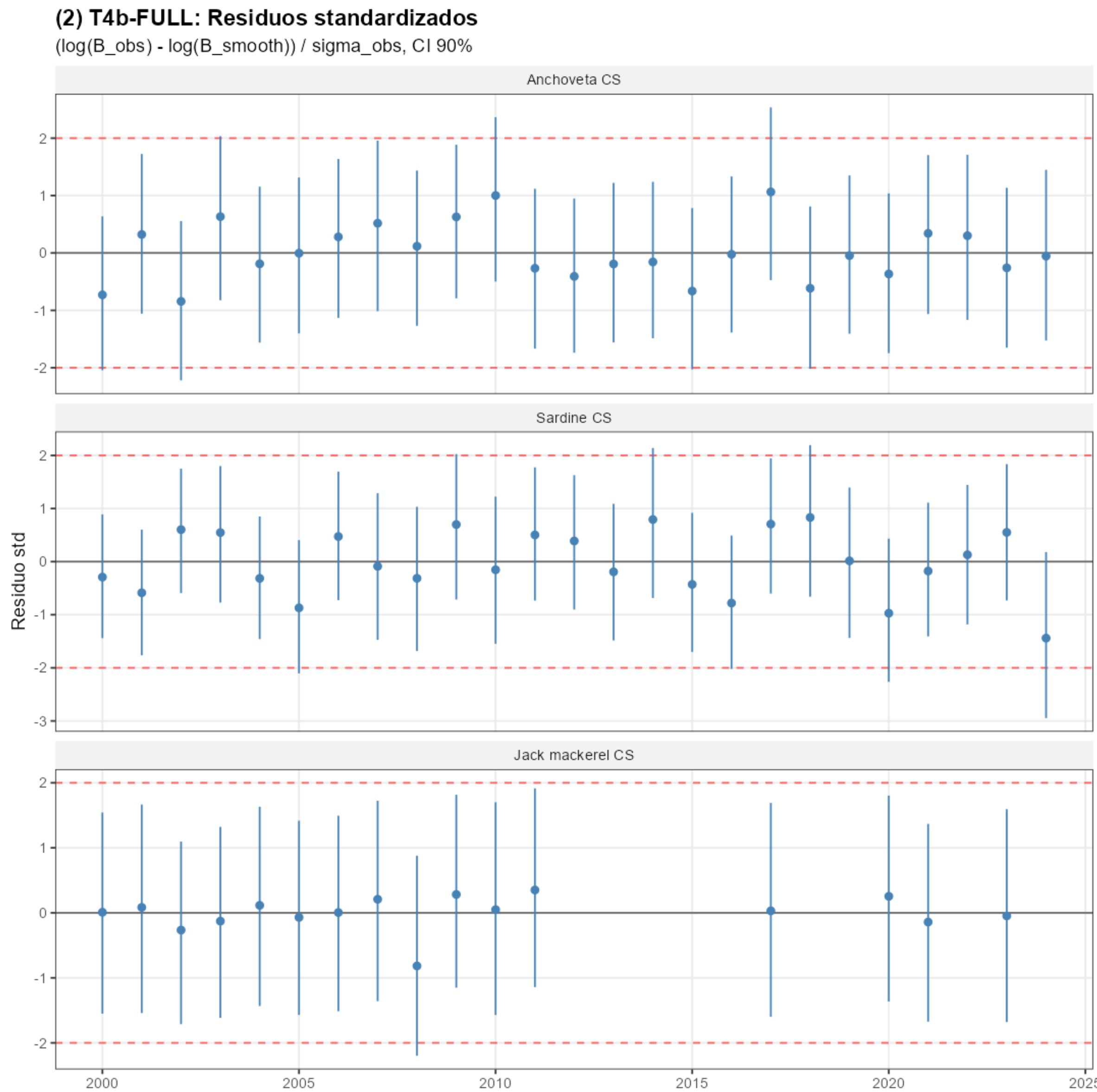

Figure C.2: Standardised year-level residuals of the full specification by stock. The Ljung–Box test of residual autocorrelation does not reject the null at conventional levels for any of the three stocks.

D Markov-chain convergence diagnostics

Table D.1 reports the Gelman–Rubin $\hat{R}$ and the bulk/tail effective sample sizes for the top-level parameters of the full state-space specification, evaluated on four independent Hamiltonian Monte Carlo chains of 2,000 post-warmup iterations each. All $\hat{R}$ values are below 1.01 and all effective sample sizes exceed 400, indicating that posterior summaries are not contaminated by between-chain divergence or by autocorrelation between adjacent draws.

Table D.1: Markov-chain convergence diagnostics for the top-level parameters of the T4b-full state-space specification.

Parameter	N	Max $\widehat{R}$	Min ESS_{bulk}	Min ESS_{tail}
r_i^0	3	1.001	8,446	10,508
K_i	3	1.000	9,789	9,777
$B_{i,0}$	3	1.001	18,919	11,844
$\sigma_{\text{proc},i}$	3	1.003	3,353	5,997
$\sigma_{\text{obs},i}$	3	1.009	1,370	936
ρ_i^{SST}	3	1.000	8,312	10,147
ρ_i^{CHL}	3	1.001	14,319	10,633
Ω (off-diagonal)	3	1.001	5,732	8,544
All top-level parameters	24	1.009	1,370	936

Note:
Diagnostics are computed from four chains of 2,000 post-warmup iterations. Within each parameter family, the table reports the worst case across the three stocks (or the three off-diagonal entries of Omega): the largest split-Rhat and the smallest bulk- and tail-effective sample sizes. The diagonal of Omega is fixed at one and is therefore excluded. The conventional thresholds are Rhat <= 1.01 and ESS >= 400 for all top-level parameters.

E Spatial robustness of the jack mackerel non-identification result

The non-identification of $(\rho^{SST}_{\text{jurel}}, \rho^{CHL}_{\text{jurel}})$ reported in the Identification of climate shifters section of the main text admits two observationally equivalent readings: that jurel productivity is uncoupled from Centro-Sur coastal forcing at the annual aggregate, or that the relevant forcing operates at a spatial scale broader than the EEZ, diluting the local-domain signal. This appendix tests four explanations for the null — insufficient power, insufficient spatial extent, insufficient biomass coverage, and a poor choice of spatial scale — and concludes that the null is structural rather than aggregation-driven within the 2000–2024 Centro-Sur record.

E.1 Identification power on the sample

Linearising the Schaefer transition in log-differences, $\Delta \log B_{i,t+1} = \alpha_i - \beta_i (B_{i,t}/K_i) + \rho_i^X X_{t-1} + \varepsilon_{i,t}$ with $\varepsilon_{i,t} \sim \mathcal{N}(0, \sigma^2_{i,\text{resid}})$ and $\sigma_{i,\text{resid}} = \sqrt{\sigma^2_{i,\text{proc}} + 2\sigma^2_{i,\text{obs}}}$, the OLS-equivalent standard error on a sample of $N = 24$ lag-one observations is $\mathrm{SE}(\hat{\rho}_i^X) \approx \sigma_{i,\mathrm{resid}}/[\mathrm{sd}(X)\sqrt{N-K}]$ with $K = 3$ regressors. The minimum shifter magnitude detectable at 80% power for a two-sided 90% credible interval is then $|\rho_i^X|_{\min} \approx 2.56\,\mathrm{SE}(\hat{\rho}_i^X)$.

Evaluated at the posterior medians of σ_{proc} and σ_{obs}, $\sigma_{i,\mathrm{resid}}$ is 0.28 for anchoveta, 0.25 for sardine, and 1.28 for jurel — the latter roughly 4.6 times the coastal-stock average and the main reason why the prior dominates the jurel posterior. Table E.1 reports $|\rho|_{\min}$ for the three stocks and three candidate shifters. The jurel envelope spans $|\rho|_{\min}$ of 1.50 on local SST, 1.32 on local log~CHL, and 0.71 on basin-scale ENSO, against an envelope of empirically plausible biological elasticities centred near 0.05–0.30 in absolute value, far below the $|\rho|_{\min}$ envelope reported here. The sample is therefore insufficiently informative to support a credible interval for jurel under any of these forcings, regardless of the true elasticity.

Table E.1: Minimum shifter magnitude $|\rho|_{\min}$ detectable at 80% power on the 2000–2024 sample. $\sigma_{\mathrm{resid}} = \sqrt{\sigma^2_{\mathrm{proc}} + 2\sigma^2_{\mathrm{obs}}}$ evaluated at posterior medians.

Stock	σ_{resid}	$\lvert\rho_{SST}\rvert_{\min}$	$\lvert\rho_{\log\mathrm{CHL}}\rvert_{\min}$	$\lvert\rho_{\mathrm{ENSO}}\rvert_{\min}$
Anchoveta	0.28	0.33	0.29	0.16
Sardina común	0.25	0.29	0.26	0.14
Jurel	1.28	1.50	1.32	0.71

Notes. Detectable magnitudes computed from $|\rho_i^X|_{\min} \approx 2.56\,\sigma_{i,\mathrm{resid}}/[\mathrm{sd}(X)\sqrt{N-K}]$ with $N = 24$, $K = 3$. $\mathrm{sd}(X)$ are sample standard deviations of the Centro-Sur EEZ-aggregated SST and log CHL anomalies and of the Niño 3.4 index over the estimation window.

E.2 Posterior identification across alternative specifications

We re-fit the state-space jurel block under four alternative shifter configurations, each designed to test one mechanical explanation for the local null. Table E.2 reports the posterior-to-prior standard deviation ratios for the jurel shifters under each specification; ratios at unity indicate no posterior updating relative to the prior.

Table E.2: Posterior-to-prior standard deviation ratios of the jurel climate shifters under four alternative specifications.

Test	Specification	$\sigma_{\text{post}}/\sigma_{\text{prior}}$ (SST)	$\sigma_{\text{post}}/\sigma_{\text{prior}}$ (log CHL o
Spatial extent	EEZ domain $\mathcal{D}_1$	0.998	1.014 (log CHI
	Offshore-extended $\mathcal{D}_2$	1.002	1.009 (log CHI
	Southeast Pacific $\mathcal{D}_3$	0.999	1.011 (log CHI
Biomass coverage	Dual-source (Centro-Sur + Northern acoustic)	0.94	0.99 (log CHL
Spatial scale	Basin-scale ENSO Niño 3.4, lag 1	—	0.98 (ENSO)
	Basin-scale ENSO Niño 3.4, lag 2	—	1.01 (ENSO)
Joint specification	Local SST + local log CHL + ENSO (lag 1)	1.03	1.00 (log CHL); 0.98

Notes. $\mathcal{D}_1$ is the Chilean EEZ between 18°S and 42°S out to 200 nautical miles. $\mathcal{D}_2$ extends to 1,000 km offshore. $\mathcal{D}_3$ is the region Pacific domain 5°S–45°S, 70°W–120°W. Spatial aggregates are area-weighted (with cos(lat)) and centred on the 2000–2024 mean. The extension augments the Centro-Sur biomass likelihood with the Northern Chilean acoustic series under a common shifter $\rho^{\text{common}}_{\text{jurel}}$, r the 0.88 log-scale cross-series correlation. The ENSO specification replaces the local coastal shifters with the Niño 3.4 index at the i The joint specification estimates all three shifters simultaneously for jurel, holding the anchoveta and sardina blocks at the main sp the Stock dynamics subsection of the main text. Anchoveta and sardina shifter ratios remain stable at 0.43–0.83 across all configu reported).

All four tests deliver a posterior-to-prior ratio between 0.94 and 1.03 for the jurel shifters, statistically indistinguishable from unity given the prior dispersion. The two ratios closest to "informative updating" — the dual-source extension at 0.94–0.99 and the basin-scale ENSO at 0.98 — remain well above the 0.7 threshold below which posterior intervals discriminate the shifter from zero on the prior support, consistent with the analytic power calculation of Table E.1.

E.3 Dual-source extension

The Centro-Sur and Northern Chilean acoustic biomass series exhibit a 0.88 log-scale correlation across overlapping years (2012–2024), motivating a joint-likelihood specification in which a common shifter ρ^{common} is identified simultaneously on both records. The extension multiplies the number of informative jurel-biomass observations by approximately 1.7 while preserving the prior structure of the main fit. The resulting $\sigma_{\text{post}}/\sigma_{\text{prior}}$ ratios of 0.94 (SST) and 0.99 (log~CHL) are the strongest posterior updating observed in any specification considered in this appendix, but remain well above the threshold for material identification.

E.4 Basin-scale ENSO test

Replacing the local coastal shifters with the basin-scale ENSO Niño 3.4 index tests whether the relevant climatic forcing for jurel operates at the basin scale rather than the coastal scale, consistent with the empirical evidence of Peña-Torres et al. (2017) (ENSO-driven shifts in jurel location choices) and Arcos et al. (2001) (SPRFMO-scale stock dynamics under broad-basin forcing). The specification estimates a single $\rho^{\text{ENSO}}_{\text{jurel}}$ on lag-1 and lag-2 ENSO anomalies in alternative runs. The resulting ratios of 0.98 (lag~1) and 1.01 (lag~2) are both indistinguishable from unity, consistent with the analytic $|\rho_{\text{ENSO}}|_{\min} = 0.71$ of Table E.1.

A prior-propagation sensitivity combines the posterior of $\rho^{\text{ENSO}}_{\text{jurel}}$ with the CMIP6 ensemble of basin-scale temperature deltas to compute a 90% predictive interval for the jurel productivity factor under SSP5-8.5 end-of-century. The interval spans approximately three orders of magnitude ($[0.05, 19.5]$), confirming that prior-propagated projections are uninformative for policy and validating the fixed-$r^{\star}_{\text{jurel}}$ convention adopted in the main text.

A separate exploratory attempt to incorporate the SPRFMO range-wide assessment (OROP-PS, encompassing Chile, Peru, Ecuador and adjacent high-seas) fails on coherence grounds: the augmented likelihood exhibits multimodal posterior support and unsatisfactory chain mixing, indicating that the broader geographic aggregate is not integrable with the Centro-Sur record under a single shifter.

E.5 Implication for the main claim

The four tests above rule out the most natural mechanical explanations for the jurel non-identification: it is not a small-sample artefact within the available power envelope (Table E.1); it does not respond to broader spatial aggregation of the local shifters (E.1: $\mathcal{D}_1$ through $\mathcal{D}_3$); it does not respond to expanded biomass coverage through the Northern Chilean acoustic series (E.2 dual-source); and it does not respond to replacing the local shifters with a basin-scale forcing (E.3

ENSO). The convergence of these four tests points to a structural rather than aggregation-driven limitation of the present record. Closing the identification gap will require either a substantially longer biomass record (the 25-year window limits any moderate-magnitude shifter to below detection) or a basin-scale assessment with explicit cross-stock spillover within a fully transboundary stock-dynamics specification. Within the present record, holding $r^{\star}_{\text{jurel}}$ fixed at the prior mean is the defensible reporting choice and is the convention adopted throughout the main text.

F Decomposition of projection uncertainty across the CMIP6 ensemble

The comparative statics of the Climate change projections section of the main text integrate two sources of uncertainty: the posterior of the structural shifters $(\rho_s^{SST}, \rho_s^{CHL})$ and the spread of climate-forced trajectories across the CMIP6 ensemble. Letting $X_{s,m,d}$ denote the percentage change in r_s implied by climate model m and posterior draw d, the law of total variance decomposes $\text{Var}_{(m,d)}(X)$ into a within-model (posterior) component $\mathbb{E}_m[\text{Var}_d(X \mid m)]$ and a between-model (climate-ensemble) component $\text{Var}_m[\mathbb{E}_d(X \mid m)]$. We compute the decomposition by Monte Carlo from the posterior (16,000 draws) crossed with the six-model ensemble of the Climate change projections section of the main text. Jack mackerel is omitted because its non-identified shifter implies a posterior dominated by the prior, leaving the decomposition without a structural reading.

The decomposition reveals a sharp contrast between the two identified stocks. For *sardine*, the between-model component dominates in three of four scenario–window pairs (77–91% of total variance): the within-model posterior on ρ_{sard}^{SST} is tight enough that the bottleneck is climate ensemble disagreement, not structural inference. The single exception is SSP5-8.5 end-of-century, where the within share rises to 59% due to a floor effect at -100% rather than a substantive shift. For *anchoveta*, the within-model component is comparable to or larger than the between-model component across all scenarios, reaching 97% under SSP5-8.5 end-of-century. The bottleneck on the anchoveta projection is therefore the structural posterior.

Table F.1: Variance decomposition of $\Delta r_s^*/r_s^{(0)}$ into within-model (posterior) and between-model (CMIP6) components. n_m is the number of CMIP6 models contributing to the cell. Jack mackerel omitted (non-identified shifter).

Stock	Scenario	n_m	Mean %Δ	SD total	% within	% between
Anchoveta CS	SSP2-4.5, 2041-2060	5	−48.0%	0.180	64%	36%
Anchoveta CS	SSP2-4.5, 2081-2100	5	−64.9%	0.232	62%	38%
Anchoveta CS	SSP5-8.5, 2041-2060	6	−58.4%	0.188	68%	32%
Anchoveta CS	SSP5-8.5, 2081-2100	6	−83.1%	0.339	97%	3%
Sardine CS	SSP2-4.5, 2041-2060	5	−77.7%	0.172	9%	91%
Sardine CS	SSP2-4.5, 2081-2100	5	−94.2%	0.064	23%	77%
Sardine CS	SSP5-8.5, 2041-2060	6	−86.7%	0.122	13%	87%
Sardine CS	SSP5-8.5, 2081-2100	6	−99.5%	0.010	59%	41%

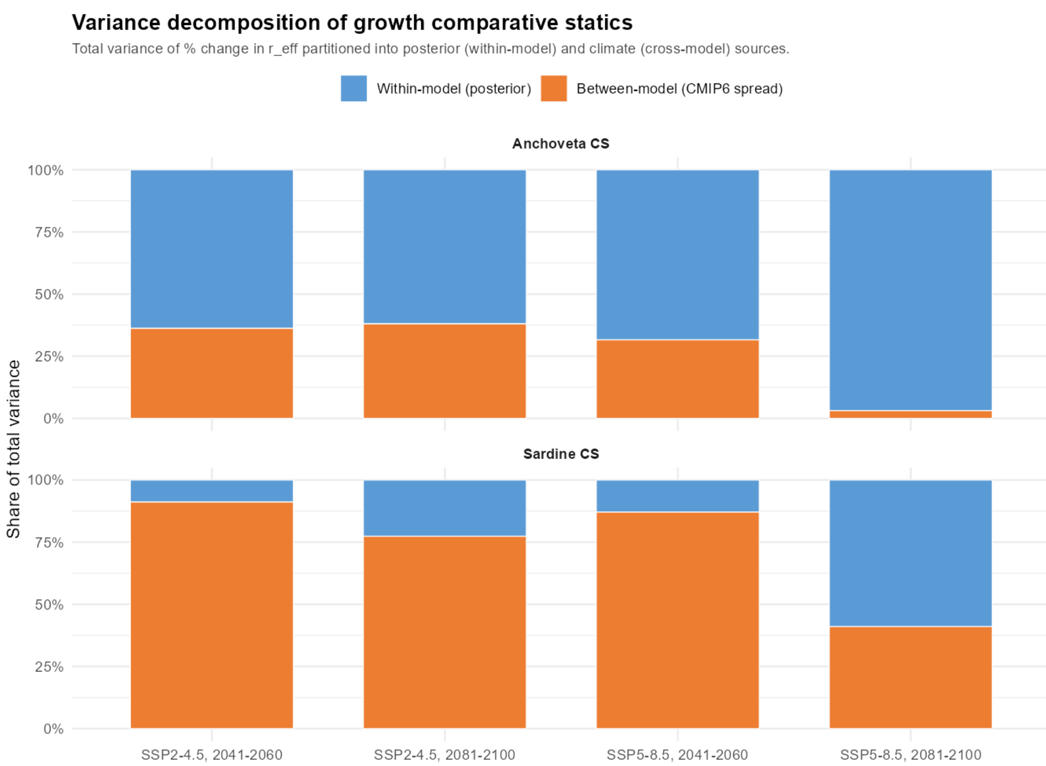

Figure F.1: Variance decomposition of $\Delta r_s^*/r_s^{(0)}$ across the CMIP6 ensemble. Bars stack to one within each scenario–window pair, splitting total variance into within-model (posterior, blue) and between-model (CMIP6 spread, orange) components. Sardine projections are climate-driven except in the SSP5-8.5 end-of-century cell where the floor effect inverts the ratio mechanically. Anchoveta projections are posterior-driven, particularly under SSP5-8.5 end-of-century (97% within). Jack mackerel omitted.

The implication for uncertainty management is species-specific: shrinking the anchoveta projection uncertainty requires a longer or more informative survey series, while the sardine projection requires either a larger CMIP6 ensemble or a defensible ensemble-weighting procedure.

G Decomposition of trip-level projection uncertainty across the CMIP6 ensemble

The fleet-level comparative statics on trips reported in the Total annual trips section of the main text (the trip comparative-statics table (main text)) integrate over three sources of uncertainty: the posterior of the structural shifters ($\rho_s^{SST}, \rho_s^{CHL}$) given the historical record, the heterogeneity of catch composition ($\omega_{v,s}$) and historical quota allocation (H_v^{alloc}) across vessels within fleet, and the spread of climate-forced trajectories across the CMIP6 ensemble. This appendix decomposes the total variance of the projected percentage change in fleet-level annual trips, $\Delta T_f / T_f^{(0)}$, into a within-model component (posterior uncertainty pooled with vessel heterogeneity within fleet) and a between-model component (CMIP6 ensemble spread), using the law of total variance. It is the companion to Appendix F and answers a different diagnostic question: *whether the narrow cross-model interquartile range observed for fleet-level trip responses in the trip comparative-statics table (main text) reflects climate consensus or a floor-effect saturation of the underlying biomass collapse.*

G.1 Decomposition

Let $X_{f,m,d,v}$ denote the percentage change in annual trips for vessel v in fleet f implied by climate model m and posterior draw d under a fixed scenario–window pair (SSP, w). The total variance of $X_{f,\cdot,\cdot,\cdot}$ across (m, d, v) admits the decomposition

$$\underbrace{\text{Var}_{(m,d,v)}(X_{f,\cdot,\cdot,\cdot})}_{\text{total}} = \underbrace{\mathbb{E}_m\left[\text{Var}_{(d,v)}(X_{f,m,\cdot,\cdot} \mid m)\right]}_{\text{within-model (posterior + vessels)}} + \underbrace{\text{Var}_m\left[\mathbb{E}_{(d,v)}(X_{f,m,\cdot,\cdot} \mid m)\right]}_{\text{between-model (climate)}} \tag{1}$$

The first term is the average variance within a model, taken over posterior draws and vessels within fleet jointly: it captures uncertainty about the structural elasticities and heterogeneity in

Table G.1: Variance decomposition of $\Delta T_f/T_f^{(0)}$ into within-model (posterior pooled with vessel heterogeneity) and between-model (CMIP6) components. n_m is the number of CMIP6 models contributing to the cell.

Fleet	Scenario	n_m	Mean %Δ	SD total	% within	% between
Artisanal	SSP2-4.5, 2041-2060	5	−18.4%	0.2010	97%	3%
Artisanal	SSP2-4.5, 2081-2100	5	−20.7%	0.2281	100%	0%
Artisanal	SSP5-8.5, 2041-2060	6	−20.3%	0.2071	100%	0%
Artisanal	SSP5-8.5, 2081-2100	6	−21.4%	0.5025	100%	0%
Industrial	SSP2-4.5, 2041-2060	5	−3.8%	0.0838	100%	0%
Industrial	SSP2-4.5, 2081-2100	5	−3.9%	0.0863	100%	0%
Industrial	SSP5-8.5, 2041-2060	6	−3.9%	0.0854	100%	0%
Industrial	SSP5-8.5, 2081-2100	6	−3.9%	0.0839	100%	0%

vessel exposure under a *fixed* climate forcing. The second term is the variance across CMIP6 models of the within-model means: it captures uncertainty about the magnitude of the climate forcing itself.

We pool posterior draws and vessel heterogeneity into the within-model component for two reasons. First, structural parallelism with Appendix F, which reports an exactly analogous two-way decomposition for stock-level intrinsic productivity. Second, vessel heterogeneity in $(\omega_{v,s}, H_v^{\text{alloc}})$ is a fixed feature of the historical fleet that does not shrink with longer historical samples or larger climate ensembles; reporting it as part of the within-model component is faithful to the reading that a longer record or a richer ensemble would not, by construction, reduce that share. A three-way decomposition that separates posterior, vessel, and climate contributions can be obtained from the same `factor_trips_dt` object generated by the trip-pipeline script and is provided in the supplementary code; the qualitative pattern is unchanged.

We compute Equation (1) by Monte Carlo from the T4b-full posterior crossed with the six-model CMIP6 ensemble, propagated through the Schaefer steady-state biomass equation under historical average fishing pressure and the negative binomial trip equation, for each of the 830 vessels in the panel.

G.2 Results

Table G.1 reports the decomposition for the four scenario–window pairs and two fleets.

The decomposition shows that the within-model component dominates the total variance across all eight scenario–window/fleet cells: it accounts for 97% of total variance for the artisanal fleet under SSP2-4.5 mid-century and saturates at 100% under SSP5-8.5 end-of-century; for the industrial fleet, the within-model share rises from 100% under SSP2-4.5 mid-century to 100% under SSP5-8.5 end-of-century. The between-model component—the share of total variance attributable to disagreement across CMIP6 models on the magnitude of climate forcing—is at most 3% of total variance in the SSP2-4.5 mid-century cells and is numerically zero, to the nearest percent, in the remaining six cells. This is *not* a statement that the CMIP6 ensemble is in tight agreement on the climate signal that drives fleet-level trips, in the sense that one would attribute to the sub-ensemble of CMIP6 models with similar equilibrium climate sensitivity; it is the mechanical signature of floor-effect saturation of the underlying biomass collapse channel.

The mechanism is straightforward. Once the Schaefer steady-state biomass factor f_v^H collapses across most posterior draws, the negative binomial trip equation maps that collapse into $\exp(\beta_f \cdot H_v^{\text{alloc}} \cdot (f_v^H - 1)) \approx \exp(-\beta_f \cdot H_v^{\text{alloc}})$, a quantity determined by the fleet-specific business mechanics (β_f and the historical quota allocation) rather than by the magnitude of the climate forcing. Once all CMIP6 models drive the system into this saturation region, the between-model spread of the within-model mean trip response collapses to a narrow band—numerically near zero in the present sample—while the within-model variance retains the dispersion induced by vessel heterogeneity in $(\omega_{v,s}, H_v^{\text{alloc}})$ and by the structural posterior across that subset of draws in which the portfolio survives. The total SD values are correspondingly small in absolute magnitude (0.5025 for the artisanal fleet and 0.0839 for the industrial fleet under SSP5-8.5 end-of-century), confirming that what little variance remains is within-model.

The means reported in Table G.1 are systematically more negative than the medians reported in the trip comparative-statics table (main text) of the main results section—for example, −21.4% for the artisanal fleet under SSP5-8.5 end-of-century in this appendix versus −17.6% in the trip comparative-statics table (main text). This is the same left-skew mechanism documented in Appendix F for the underlying productivity response: posterior draws and vessels in which the portfolio collapses pull the median up toward the floor at $\exp(-\beta_f \cdot H_v^{\text{alloc}}) - 1$, while the mean integrates over the long left tail of vessel-and-draw configurations in which the collapse drives f_v^H deep into negative territory and the trip response down with it. Both summaries are valid; we report cross-model medians in the main results table because the median is robust to the extreme tail and is the natural complement to the cross-model inter-quartile range, and we report cross-model means in this appendix because the law of total variance in Equation (1) is constructed on means rather than medians.

This reading is consistent with the underlying biomass-level decomposition reported in Appendix F: for sardine—the stock that drives the artisanal fleet's portfolio collapse—the SSP5-8.5 end-of-century cell exhibits a total SD of 0.010 and an inverted within/between ratio that the appendix explicitly reads as a floor effect at the lower bound of -100%. The fleet-level decomposition reported here inherits that floor effect: the total SD for the artisanal fleet under SSP5-8.5 end-of-century is 0.5025 and the corresponding industrial cell is 0.0839, both of which are small absolute magnitudes and mechanically consistent with the saturated regime.

The substantive policy implication is that, for fleet-level trip projections, the marginal information value of expanding the CMIP6 ensemble is small relative to that of tightening the structural posterior or refining the description of vessel heterogeneity within fleet. The dominant uncertainty is *not* "which climate model is right" but "given that all of them imply portfolio collapse for the two coastal stocks, how does the resulting heterogeneous economic exposure across vessels translate into trip-level effort." This is the econometric counterpart of the well-known result in the climate-impacts literature that ensemble averaging masks first-order distributional heterogeneity downstream of the climate signal (Cline et al. 2017; Free et al. 2019; Kasperski and Holland 2013; Oken et al. 2021).

Figure G.1 visualises the decomposition.

H Permit portfolios and sectoral counterfactual

This appendix reports three pieces of evidence that underpin the institutional mechanism developed in the Discussion section of the main text. Section H.1 documents the revealed multi-species portfolio held by each sector, refuting the common reading that the artisanal–industrial asymmetry estimated in the main text is driven by single-species permit rigidity at the vessel level. Section H.2

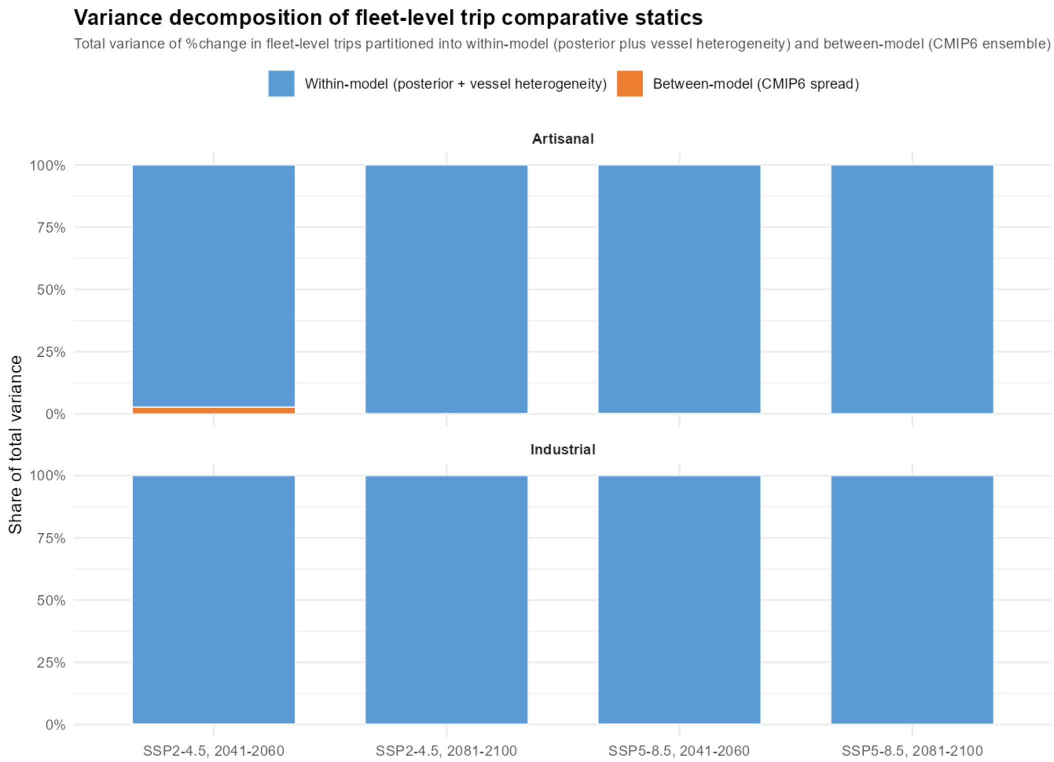

Figure G.1: Variance decomposition of $\Delta T_f / T_f^{(0)}$ across the CMIP6 ensemble. Bars stack to one within each scenario–window pair and split the total variance into within-model (posterior pooled with vessel heterogeneity, blue) and between-model (CMIP6 spread, orange) components. The within-model component dominates across all cells, reflecting the floor-effect saturation of the underlying Schaefer steady-state biomass collapse rather than tight cross-model agreement on the climate signal.

computes a policy counterfactual in which the projected SSP5-8.5 end-of-century catch allocation is recomputed under the 2025 fractioning reform (Congreso Nacional de Chile 2025), in force through 2040, and contrasts it with the pre-reform regime that prevailed during the estimation window. Section H.3 documents the realised inter-sectoral quota cession flow over 2013–2024, showing that the industrial sector already cedes the bulk of its assigned small-pelagic TAC to artisanal operators, so that the de jure 2025 reform of Section H.2 should be read as an upper bound on the post-cession allocation.

H.1 Multi-species permit portfolios in both sectors

A natural concern with the institutional reading of our main results is that artisanal vessels may operate under more restrictive permit portfolios than industrial vessels, in which case the headline asymmetry would reflect within-vessel rigidity rather than the cross-sector wedge. Two complementary data sources allow us to test this directly. For the industrial sector we use the registry of vessel-level participation coefficients (*coeficientes de participación relativa*) for jack mackerel, sardine and anchoveta, in force through 2012, released by SUBPESCA in response to the Ley de Transparencia request AH002T-0007379 (Subsecretaría de Pesca y Acuicultura (SUBPESCA) 2026). This is the last pre-LTP snapshot of authorised industrial portfolios and remains the legal basis for the Class A LTP coefficients in force thereafter. For the artisanal sector, the equivalent administrative registry of permits by RPA and species is not publicly available; we therefore use the 2001–2024 IFOP logbook (454,308 trip records on 1,583 vessels covering 81 species) and treat the set of species each vessel *actually landed* as a revealed-preference lower bound on the species for which it holds operational authorisation, since vessels cannot legally land species outside their authorised portfolio.

Table H.1 reports four portfolio-breadth indicators by sector bucket, with vessel types following the IFOP scientific-observer typology (Vega Muñoz et al. 2016): the mean number of distinct species landed over the panel lifetime, the share of vessels landing two or more species over that lifetime, the mean number of distinct species landed per vessel-year, and the share of vessel-years with two or more species active.

Table H.1: Permit-portfolio breadth indicators by sector bucket, Chile small pelagic logbook 2001–2024 (artisanal) and SUBPESCA 2012 registry (industrial).

Bucket	n vessels	Sp. lifetime	Multi-sp. lifetime (%)	Sp. per year	Multi-sp. per year (%)
Industrial (IND)	129	7.63	87.6	4.14	96.9
Lancha Menor (LM)	888	6.27	89.8	3.81	89.9
Unclassified artisanal	367	4.76	89.6	3.46	91.3
Bote Menor (BM)	189	2.44	63.0	2.14	63.4

Notes. IND row: SUBPESCA 2012 registry (1,149 authorisations across 299 vessels; 75.6% hold the full anchoveta–sardine–jack mackerel triplet, 67.6% are authorised in ≥ 2 macrozones). Artisanal rows: 2001–2024 IFOP scientific-observer logbook, lances with valid species codes. Buckets follow the IFOP 2016 typology used in the main text: LM = *Lancha a Motor*, BM = *Bote a Motor*, industrial reported as PAM; LM and BM separate empirically by hold capacity (median 64 vs 4 m^3, non-overlapping IQRs). The anchoveta–sardine–jack mackerel triplet accounts for 80–86% of artisanal and 67% of industrial lances.

Three substantive readings of Table H.1 support the institutional argument developed in the Discussion section of the main text. First, the count of permits per vessel does not distinguish the two sectors in a way that would generate the headline asymmetry: 87.6% of industrial vessels and 89.8% of the dominant Lancha Menor sub-bucket of the artisanal sector hold authorisations for two or more species over the panel lifetime, with similar shares per vessel-year. Second, the within-artisanal heterogeneity that anchors the LM–BM contrast in the within-artisanal TIPO_EMB table (main text) is *not* explained by greater species diversification in BM: BM in fact reports the narrowest logbook portfolio (mean 2.44 species lifetime, 2.14 per year), so the smaller projected response of BM to climate stress cannot be a permit-diversification story. The most plausible alternative is that BM operates substantially outside the regulated logbook (coastal multi-species and benthic activity), so the projected trip elasticity applies to a smaller fraction of BM's revealed livelihood than of LM's. Third, the composition of the multi-species portfolio differs sharply: 80–86% of artisanal lances and 67% of industrial lances fall on the anchoveta–sardine–jack mackerel triplet, but within that triplet artisanal activity concentrates on the anchoveta–sardine pair while industrial activity is more evenly spread, with jack mackerel and caballa together absorbing $\sim$ 70% of industrial lances. Under climate forcing, the anchoveta–sardine pair responds with the identified negative semi-elasticities of the Stock dynamics subsection of the main text, while jack mackerel productivity is non-identified at the Centro-Sur scale (Appendix E); the asymmetric concentration of the artisanal portfolio in the anchoveta–sardine pair therefore translates the same climate forcing into a sharper effective reduction of the artisanal portfolio than of the industrial portfolio, conditional on the institutional inability to recompose access across the sectoral boundary.

H.2 Policy counterfactual under the 2025 reform

The estimation window of the present paper (2018–2024) coincides with the pre-reform fractioning regime. The 2025 fractioning reform (Congreso Nacional de Chile 2025) fixes a new schedule of artisanal–industrial fractions in force through 2040. Across the Valparaíso–Los Lagos macrozone studied here, the law raises the artisanal share to 90% in anchoveta and sardine (from 78%) and partitions jack mackerel into a 30% artisanal share in Valparaíso–Los Ríos and 15% in Los Lagos (from a uniform 10% pre-reform); the fractioning table (main text) reports the two regimes side by side. A static plug-in evaluation of the climate-conditional artisanal landing under each regime is reported in the Implications of the 2025 fractioning reform subsection of the main text (the static counterfactual table (main text)); this appendix records the formal derivation and the posterior-propagated, by-model version that complements that headline table.

Letting Q_s^0 denote the 2024 Cuota Global de Captura for stock s and $\Delta r_s^\star / r_s^0$ the projected productivity change at SSP5-8.5 end-of-century, the artisanal absolute landing under fractioning regime R is

$$L_{\text{art}}^{(R)} \;=\; \sum_s \alpha_s^{(R)} \cdot \left(1 + \Delta r_s^\star / r_s^0\right) \cdot Q_s^0, \tag{2}$$

where $\alpha_s^{(R)}$ is the artisanal fraction of stock s under regime $R \in \{\text{pre-reform}, \text{2025 reform}\}$ and the sum runs over the four (stock × macrozone) units of the fractioning table (main text): anchoveta Valparaíso–Los Lagos, sardine Valparaíso–Los Lagos, jack mackerel Valparaíso–Los Ríos, and jack mackerel Los Lagos. Equation (2) reads the productivity change as a proportional adjustment to the steady-state landed volume and abstracts from any endogenous TAC-setting response. The plug-in version reported in the main text fixes $\Delta r_s^\star / r_s^0$ at the cross-model posterior median from the stock productivity comparative-statics table (main text); the posterior-propagated version reported below integrates explicitly over the CMIP6 six-model ensemble and the within-posterior 90% envelope, returning cross-model median plus inter-quartile range for the central estimate and the median within-posterior 90% CI for the structural uncertainty axis, exactly as in the trip comparative-statics table (main text).

Table H.2: Posterior-propagated artisanal absolute landing under the pre-reform and 2025 fractioning regimes, SSP5-8.5 end-of-century, Centro-Sur (Valparaíso–Los Lagos), 2024 Cuotas Globales.

Stock	Regime	Cross-model [IQR] (kt)	Within-post. CI (kt)
Anchoveta	Pre-reform	17.1 $[14.0, 20.8]$	$[2.9, 75.5]$
Anchoveta	2025 reform	19.7 $[16.2, 24.0]$	$[3.3, 87.2]$
Sardine	Pre-reform	0.3 $[0.1, 1.4]$	$[0.1, 1.7]$
Sardine	2025 reform	0.3 $[0.1, 1.6]$	$[0.1, 1.9]$
Jack mackerel	Pre-reform	65.9 $[65.9, 65.9]$	$[65.9, 65.9]$
Jack mackerel	2025 reform	185.7 $[185.7, 185.7]$	$[185.7, 185.7]$
Total	Pre-reform	83.3 $[80.0, 88.1]$	$[68.8, 147.4]$
Total	2025 reform	205.7 $[201.9, 211.3]$	$[189.1, 279.7]$

Notes. Cross-model statistics across six CMIP6 models of the per-model posterior median; within-posterior 90% CI is the median across models of per-model $[q_{05}, q_{95}]$. Jack mackerel is held at $\Delta r^\star / r^0 = 0$ (non-identified at Centro-Sur scale).

Three readings of the posterior-propagated counterfactual sharpen the plug-in headline reported in the main text. First, the cross-model inter-quartile range is narrow at the aggregate level (a few kilotonnes on either side of the 122 kt median total), reflecting the floor-effect saturation documented in Appendix G: once the small-pelagic productivity collapses, the cross-model dispersion of L_{art} is small in absolute terms because the residual base on which it operates is small. Second, the within-posterior 90% CI is substantially wider for anchoveta than for sardine, because the posterior on $(\rho^{SST}_{\text{anch}}, \rho^{CHL}_{\text{anch}})$ admits a wider envelope of climate-conditional productivity factors than the sardine pair. Third, the jack mackerel contribution to the artisanal landing is invariant across both axes of uncertainty—it appears as a point in the table—because the non-identification convention holds $(1 + \Delta r^\star_{\text{jack mackerel}} / r^0_{\text{jack mackerel}}) = 1$; relaxing this convention to the ENSO-prior-propagated envelope of Appendix E.4 would add a third source of uncertainty whose magnitude is bounded by the three-orders-of-magnitude range reported there. The plug-in headline in the main text is therefore robust to the cross-model and within-posterior axes that the appendix makes explicit, and the institutional reading developed in the Discussion section of the main text carries over unchanged.

H.3 Inter-sectoral cession evidence from SERNAPESCA quota control reports

This subsection documents the realised inter-sectoral cession flow, the single administrative reallocation channel admitted by the de jure fractioning regime. The data source is the annual SERNAPESCA quota control report, published as an Excel workbook for each pair (anchoveta–sardine Valparaíso–Los Lagos) and (jack mackerel Arica y Parinacota–Los Lagos). For each species–year we extract the assigned, ceded, effective, and captured tonnage by sector from the workbook's summary sheet.

Two qualitative facts emerge from the consolidated 2013–2024 series, which we report in two tables separated by species pattern: anchoveta and sardine in Table H.3 and jack mackerel in Table H.4. First, in the anchoveta and sardine fisheries the industrial sector cedes a large share of its assigned TAC to artisanal cessionaries in every year of the 2013–2024 window: the share of the industrial assignment transferred to artisanal operators ranges from 12% to 72% over 2013–2017 and tightens to 85%–∼100% over 2018–2024. The cession volume scales with the TAC and, in the 2018–2024 subperiod, ranges from 26 to 50 kt per year in anchoveta and from 51 to 77 kt per year in sardine. Second, in the jack mackerel fishery the cession flow is an order of magnitude smaller relative to the assigned TAC—never exceeding 14% of the industrial assignment in any year of the 2013–2024 window and falling below 5% in eight of the twelve years—and reverses direction across years: artisanal-to-industrial cessions dominate the 2013–2014 and 2016–2022 subperiods, while industrial-to-artisanal flows dominate the 2015 and 2023–2024 rows of Table H.4.

The implication for the main-text policy interpretation is twofold. First, the structural asymmetry that the paper identifies is estimated on realised landings, which are post-cession; the asymmetry therefore reflects the post-cession allocation produced by the joint pre-reform fractioning and cession regime rather than the de jure fractioning alone. Second, the static counterfactual in the Implications of the 2025 fractioning reform subsection of the main text and in equation (2) of this appendix should be read as a de jure upper bound on the post-cession allocation: because the industrial sector already cedes the bulk of its assigned small-pelagic TAC to artisanal operators under the pre-reform regime, the 12 percentage-point reduction in the industrial de jure share under the 2025 reform translates into a proportionally smaller change in realised artisanal landings. Cession contracts under the inter-sectoral channel are bilateral agreements between LTP-holders and

RPA-holders subject to ex-ante SUBPESCA authorisation. The post-cession allocation therefore depends on the de jure fractioning combined with the realised cession flows documented in Tables H.3 and H.4.

Table H.3: Inter-sectoral cession flow, anchoveta and sardine, Valparaíso to Los Lagos, 2013–2024. Source: SERNAPESCA quota control reports.

Year	Species	IND assigned (kt)	IND cession (kt)	Share ceded
2013	anchoveta	25.5	−3.1	12%
2013	sardine	128.5	−31.1	24%
2014	anchoveta	7.7	−3.9	51%
2014	sardine	104.8	−46.6	44%
2015	anchoveta	6.3	−2.5	40%
2015	sardine	65.2	−21.8	33%
2016	anchoveta	7.3	−2.1	29%
2016	sardine	59.8	+15.6	26%
2017	anchoveta	12.6	−9.0	72%
2017	sardine	72.4	−48.9	67%
2018	anchoveta	12.8	−10.7	84%
2018	sardine	74.2	−63.2	85%
2019	anchoveta	27.4	−25.8	94%
2019	sardine	76.0	−69.6	92%
2020	anchoveta	38.6	−33.3	86%
2020	sardine	67.2	−60.9	91%
2021	anchoveta	45.3	−45.1	∼100%
2021	sardine	76.9	−76.5	∼100%
2022	anchoveta	52.2	−50.5	97%
2022	sardine	77.7	−71.6	92%
2023	anchoveta	39.9	−37.1	93%
2023	sardine	63.3	−61.7	98%
2024	anchoveta	53.8	−45.5	85%
2024	sardine	53.9	−51.0	95%

Notes. Industrial share of Cuota Global de Captura, Valparaíso to Los Lagos. Negative cession denotes industrial-to-artisanal flows.

Table H.4: Inter-sectoral cession flow, jack mackerel, 2013–2024. Source: SERNAPESCA quota control reports.

Year	Species	IND assigned (kt)	IND cession (kt)	Share ceded
2013	jack mackerel	138.4	+0.5	0%
2014	jack mackerel	198.6	+27.3	14%
2015	jack mackerel	203.1	−17.2	8%
2016	jack mackerel	203.0	+11.7	6%
2017	jack mackerel	228.3	+9.7	4%
2018†	jack mackerel	326.9	+16.1	5%
2019†	jack mackerel	339.0	+12.0	3%
2020†	jack mackerel	1268.1	+11.8	1%
2021†	jack mackerel	448.7	+13.3	3%
2022†	jack mackerel	516.4	+15.2	3%
2023†	jack mackerel	637.0	−0.7	0%
2024†	jack mackerel	728.6	−31.5	4%

Notes. Conventions as in Table H.3. †2018–2024 is the national aggregate (the SERNAPESCA summary sheet is not disaggregated by sub-macrozone); the share-ceded column remains comparable because the additional Arica-to-Atacama volume enters both numerator and denominator.

I Limitations and caveats

This appendix states in full the seven caveats summarised in the Discussion section of the main text and qualifies the implication of each for the interpretation of the main results.

Several caveats apply to the present analysis. First, the projections hold ex-vessel prices and management rules constant at historical levels, implying that the long-run responses capture the mechanical effect of changed climate forcings without accounting for endogenous price adjustments or adaptive management. Sumaila et al. (2011) note that reduced fish supply under climate change could increase prices, partially offsetting revenue losses from lower catches, while Lam et al. (2016) project sizable changes in global fisheries revenues by the 2050s once price responses and cross-country trade are accounted for. Incorporating an inverse demand system and numerical optimization of quota paths is a natural extension to capture these equilibrium adjustments. Second, although our projections integrate over a six-model CMIP6 ensemble, the cross-model spread in chlorophyll-a remains substantial: the cross-model median of $\Delta \log \mathrm{CHL}$ is close to zero in all scenario–window pairs while the inter-model dispersion is comparable in magnitude to the median, reflecting the well-documented disagreement among Earth system models on the direction and magnitude of primary productivity changes (Sumaila et al. 2011). The variance decomposition reported in Appendix F shows that, for the sardine projection, the between-model component accounts for 77–91% of total projection variance under three of the four scenario–window pairs, so the residual ensemble uncertainty on this stock is large despite the multi-model integration. The fleet-level trip projections in the trip comparative-statics table (main text), by contrast, exhibit narrow cross-model interquartile ranges that should not be read as climate consensus: the corresponding decomposition reported in Appendix G attributes 97–100% of the total variance in $\Delta T_f / T_f^{(0)}$ to the within-model component (posterior pooled with vessel heterogeneity within fleet), reflecting a floor-effect saturation in which the stock-dynamics steady-state collapse drives the trip response into a fleet-specific plateau independent of the climate magnitude. The substantive implication is that the marginal information value of expanding the CMIP6 ensemble for fleet-level trip projections is small relative to that of tightening the structural posterior on $(\rho_s^{SST}, \rho_s^{CHL})$ or refining the description of vessel heterogeneity in $(\omega_{v,s}, H_v^{\mathrm{alloc}})$; what governs the heterogeneity in trip outcomes across vessels and fleets is not which CMIP6 model is correct—given that all of them imply portfolio collapse for the two coastal stocks—but the differential economic exposure to that collapse imposed by historical catch composition and quota allocation. Third, the shifters are identified on a 25-year time series and, while the state-space specification propagates process-noise and measurement-noise uncertainty into the posterior, its ability to detect non-linearities in the climate response is limited. The posterior-to-prior standard-deviation ratios reported in the climate-shifter posterior identification table (main text) provide a direct empirical check that the data update the prior for anchoveta and sardina común (ratios in the 0.43–0.83 range), while leaving the jurel shifters at the prior (ratios essentially equal to one)—the latter being the basis of our non-identification finding. Fourth, our models do not explicitly incorporate risk preferences or production risk, which may play a role in shaping fleet-level responses to environmental variability (Kasperski and Holland 2013; Sethi et al. 2014). Fifth, the biological-closure variable in the trip equation is coded as a zonal proxy with no within-year temporal variation, set to the nominal regulatory ceiling of the reproductive closure regime under the relevant SUBPESCA decrees: the V-VIII zone receives 151 days and the IX-XIV zone receives 182 days throughout the panel. The underlying regulatory chronology is in fact non-trivial—D.Ex. 115/1998 governed the closure regime through July 2016 with annual modifications (D.Ex. 1661/2009 establishing 21 August to 21 October as the standing period, with year-specific adjustments in D.Ex. 705/2010, 796/2012, 747/2013, and 598/2015); D.Ex. 530/2016 then introduced a referential period of 6 July to 31 October with a fixed core (3 August to 4 October) and dynamic windows triggered by weekly IGS/PHA biological indicators; D.Ex. 137/2021 refined the indicator thresholds; and D.Ex. 05/2024 superseded the 530/2016 decree—and the actual realised closure days reported by the SUBPESCA Indicadores Biológicos portal range from 98 (in 2019) to 230 (in 2024) across regions and years. Constructing a year-by-year closure variable that reconciles the nominal regulatory regime with the actual realised closure days reported in the portal (available from 2019 onward) and with auxiliary recruitment-closure decrees not regulated by the reproductive-closure chain documented above is a non-trivial data-engineering task that we leave to follow-up work; for the present paper the zonal proxy captures the structural V-VIII / IX-XIV differential (31 days) that, under the year-fixed-effects specification of the Total annual trips section of the main text, is the variation that identifies β_{closure}. Sixth, the trip equation panel is constructed from IFOP logbook records and is therefore representative of the purse-seine fleet that operates under the LTP quota architecture; smaller artisanal vessels using non-purse-seine gears (lampara, lines) operate outside the logbook system. Cross-validation against the SERNAPESCA vessel-level landings database confirms that the panel's portfolio weights $\omega_{v,s}$ match the SERNAPESCA aggregate within two percentage points for the industrial fleet and within three percentage points (catch-weighted) for the purse-seine artisanal subset that the panel represents. The non-purse-seine artisanal segment, which accounts for approximately 30% of total Centro-Sur artisanal landings and exhibits a higher proportional exposure to jack mackerel, is not modeled here; a sample-weighting extension that incorporates that segment is left to future work. The biomass likelihood, by contrast, is constructed from SERNAPESCA all-gear landings and is unaffected by this restriction. Seventh, the direct weather channel is constructed from five of the six CMIP6 ensemble members because CESM2 does not publish near-surface wind components (uas, vas) in the CMIP6 DRS for the relevant scenarios and time windows; CESM2 contributes to the indirect biomass channel through SST and chlorophyll deltas (six members in SSP5-8.5, five in SSP2-4.5 by the chlorophyll restriction noted in the Projection approach subsection of the main text) but enters the direct channel with $\Delta\text{wind} = 0$ by construction. The contribution of CESM2 to the cross-model spread of fleet-level trips therefore flows entirely through the indirect channel, which is the dominant source of variance in any case under the floor-effect saturation documented in Appendix G.

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